CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

I Semester of BSc+M Sc Dual Degree Examination March, 2018

BS104 BASIC MATHEMATICS FOR BIOLOGISTS

Date:29/03/2018,Thursday Time:10:00 a.m. to 10:20 a.m. Maximum Marks:10

MCQ

Important Instructions:

- Tick the correct answer and it should be written in question paper itself.
- Use of non-programmable calculator is allowed.

Q - I Choose the correct answer for the following questions.

1. If $A = \{3, 5\}$ and $B = \{1, 2, 3\}, C = \{3, 4\}$ then $A \cup B \cup C =$ ______.

   (i) $\{1, 2, 3, 4, 5\}$ (ii) $\{3\}$ (iii) $\{3, 4\}$ (iv) ϕ

2. If $f(x) = x^2$ then $\int x^2 dx =$ ______ ,

   (i) $2x + C$ (ii) $\dfrac{x^3}{3} + C$ (iii) $\dfrac{x^3}{2} + C$ (iv) $\dfrac{x^2}{3} + C$

   where C is any arbitrary constant.

3. For the function $f(x) = 2x - 4$, then______.

   (i) $\lim_{x \to 1} f(x) = 6$ (iii) $\lim_{x \to 1} f(x) = -2$

   (ii) $\lim_{x \to 1} f(x) = 2$ (iv) $\lim_{x \to 1} f(x) = -6$

4. Let f denote a function from A into B, $f : A \to B$. Then range of f is______.

(i) A
(ii) B
(iii) subset of B
(iv) None of these

5. If $y = \sin bx$, then derivative of y is______.

(i) $-\cos bx$ (ii) $-b\cos bx$ (iii) $b\sin bx$ (iv) $b\cos bx$

6. The degree of polynomial $3x + 2$ is______.

(i) 2 (ii) 1 (iii) 3 (iv) 5

7. If $\begin{vmatrix} 5 & 1 \\ 0 & x \end{vmatrix} = 10$ then $x =$______.

(i) 2 (ii) 3 (iii) $-\frac{2}{3}$ (iv) 7

8. For two matrices $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 2 \\ 0 & 0 \end{bmatrix}$, $A + 2B =$______.

(i) $\begin{bmatrix} 3 & 6 \\ 3 & 4 \end{bmatrix}$
(ii) $\begin{bmatrix} 1 & 2 \\ 0 & 0 \end{bmatrix}$
(iii) $\begin{bmatrix} 3 & 5 \\ 7 & 6 \end{bmatrix}$
(iv) $\begin{bmatrix} 1 & 2 \\ 4 & 2 \end{bmatrix}$

9. A 40g of solution is equally distributed in 8 test tubes then how much solution is equal to 15 g?

(i) $\frac{3}{8}$ (ii) $\frac{15}{8}$ (iii) $-\frac{5}{8}$ (iv) $\frac{5}{8}$

10. The quadratic equation $x^2 - 5x - 6 = 0$ has ______.

(i) two different real roots
(ii) two equal real roots
(iii) two different complex roots
(iv) None of these

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Date:29/03/2018,Thursday Time:10:20 a.m. to 12:00 noon Maximum Marks:25

Important Instructions:

- Section I and II must be attempted in SINGLE ANSWER SHEET.
- Make suitable assumptions and draw neat figures wherever required.
- Use of non-programmable calculator is allowed.
- Show necessary calculations.

SECTION-I

Q-I Answer the following questions.(Any Two) [10]

(i) Let $U = \{1, 2, 3, 4, 5, 6, 7, 8, 9\}$ and suppose that $A = \{1, 2, 3, 4, 5\}$, $B = \{4, 5, 6, 7\}$ and $C = \{5, 6, 7, 8, 9\}$ then find (a)A' (b) $B \cup C$ (c)$A' \cup B$ (d)$B' \cap C$

(ii) Tabulated the value of x^2 in $[-2, 2]$ and plot the graph of the function.

(iii) Find the root of the quadratic equations (a)$m^2-8m+16 = 0$ (b)$m^2-2m-3 = 0$.

SECTION-II

Q-II Answer the following questions.(Any Five) [15]

(i) What is the %(v/v) concentration of a solution of ethanol made by mixing 25 ml ethanol with 225 ml of water?

(ii) Solve by Cramers Rule

$3x + 4y - 3z = 5, \qquad 3x - 2y + 4z = 7 \qquad 3x + 2y - z = 3$

(iii) Find the following limits (a)$\lim\limits_{x\to 1} \dfrac{1 + x + x^2}{7x}$ (b)$\lim\limits_{x\to 4} \dfrac{x^2 - 16}{x + 4}$.

(iv) Calculate the value of the determinate $\begin{vmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{vmatrix}$

(v) Differentiate $y(x)=log(7x) - 5 + x^3 - \frac{2}{5}e^x + sin(5x) + x^{-2}$ with respect to x.

(vi) Obtain integration of the function $sin3x + e^{2x} + x^6 + \dfrac{1}{x^2 + 1} + 17 + cosx$.
